



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to BAKYA LAKSHMI FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

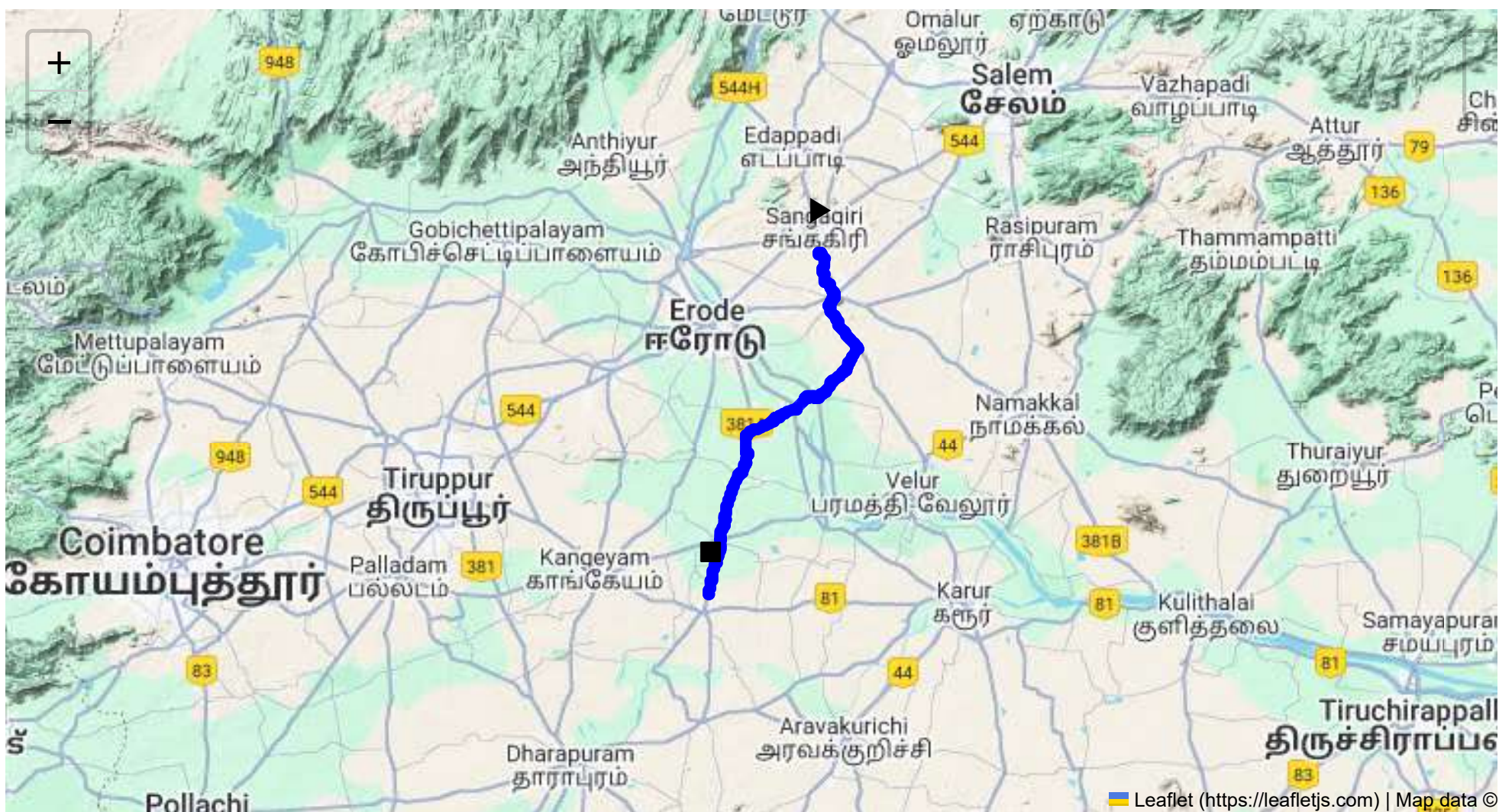
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 67.16 km**  
**Estimated Duration: 1.6 hours**  
**Adjusted Duration (Heavy Vehicle): 1.9 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.969499, 77.718803)**



## Welcome to the Journey Risk Management Study

### Route Map Overview

The route from Sangagiri to Vellakoil via Sullipalayam spans approximately 67.16 kilometers, primarily along state highways. This route is a mixture of SH 86 and SH 84A, passing through small towns and rural areas. The terrain is relatively flat, with the road running parallel to agricultural lands, small local businesses, and residential zones.



# Weather Conditions and Hazards

## Typical Weather Conditions:

- **Summer (March to May):** Hot and dry with temperatures reaching up to 40°C.
- **Monsoon (June to September):** Medium to heavy rainfall can lead to slippery roads and reduced visibility.
- **Winter (October to February):** Mild temperatures with minimal precipitation.

## Weather-Related Hazards:

- **Monsoon flooding:** Potential for short-term road closures or diversions due to waterlogging.
- **Heat effects during summer:** Increased risk of road degradation and potential for tire blowouts.

# Traffic Patterns

## Peak Hours:

- **Morning:** 7:00 AM - 9:00 AM
- **Evening:** 5:00 PM - 7:00 PM

## Congestion-Prone Areas:

- Entry and exit points near towns such as Sangagiri and Sullipalayam.
- Local markets causing slow-moving traffic, particularly near Sullipalayam.

# Road Quality and Infrastructure

- **Overall Condition:** Generally good, with occasional potholes and rough patches.
- **Infrastructure:** Presence of narrow bridges and infrequent traffic signals.
- **Maintenance:** Regular maintenance observed, but localized roadworks might cause temporary diversions.

# Emergency Alternative Routes

1. **Via NH544 to Tiruppur:** Provides a major highway alternative, though longer in distance.
2. **Through Namakkal, connecting via SH95:** A viable option if SH 86 is congested or under maintenance.

# Local Regulations for Hazardous Materials

- **Speed Limits:** Strict enforcement of 60 km/h for heavy vehicles carrying hazardous materials.
- **Transport Permits:** Mandatory for all hazardous materials.
- **Prohibitions:** Certain materials may be prohibited during festivals or local events.

# Historical Incidents

- **Common Issues:** In past, minor incidents have included tire blowouts and minor collisions mostly during monsoon.
- **Notable Incidents:** No significant major hazardous materials incidents reported recently in this route.

## Environmental and Sensitive Areas

- **Agricultural Lands:** Careful driving required to prevent spillage and contamination.
- **Water Bodies:** Rivers and local water storage areas should be noted for potential contamination risks.

## Communication Coverage

- **Overall Coverage:** Good coverage, with potential dead zones in rural stretches between Sullipalayam and Vellakoil. Notable issues might arise during severe weather.
- **Providers:** Reliance Jio, Airtel, and BSNL have intermittent coverage, especially in remote areas.

## Emergency Response Times

- **Estimated Response Times:**
  - **Sangagiri to Sullipalayam:** Approximately 20-30 minutes during peak hours.
  - **Sullipalayam to Vellakoil:** Approximately 40-50 minutes; additional delays during monsoon.

## Overall Risk Assessment Summary

This route, while generally safe, does present specific challenges such as weather-induced hazards during monsoon, minor road quality issues, and peak hour congestion. Given the infrequent but notable historical incidents, it remains essential for operators to adhere to local transport regulations, maintain communication with central dispatch, and take precautions during adverse weather conditions. With proper planning and preparedness, especially awareness of alternative routes, the risks can be managed effectively for trucks carrying hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Roundabout | High       | 11.04637, 77.73761 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.44022, 77.87546 | 15 KM/Hr    |
| 5 | Turn       | High       | 11.37868, 77.89498 | 15 KM/Hr    |
| 6 | Turn       | Medium     | 11.37842, 77.89452 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 7  | Turn       | Medium     | 11.30869, 77.92911 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.30849, 77.92910 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.24172, 77.87357 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.24000, 77.87363 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.24053, 77.86902 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.24115, 77.86891 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.24122, 77.86883 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.24159, 77.86874 | 15 KM/Hr    |
| 15 | Turn       | High       | 11.24170, 77.86842 | 15 KM/Hr    |
| 16 | Turn       | High       | 11.24248, 77.86829 | 15 KM/Hr    |
| 17 | Turn       | High       | 11.24257, 77.86776 | 15 KM/Hr    |
| 18 | Blind Spot | Blind Spot | 11.24384, 77.86689 | 10 KM/Hr    |
| 19 | Turn       | Medium     | 11.24101, 77.86034 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.24103, 77.86020 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.24200, 77.85882 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.24196, 77.85856 | 30 KM/Hr    |
| 23 | Turn       | High       | 11.18856, 77.77110 | 15 KM/Hr    |
| 24 | Turn       | Medium     | 11.16404, 77.77379 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.16377, 77.77421 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.16354, 77.77435 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.13664, 77.76524 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.04718, 77.73742 | 30 KM/Hr    |
| 29 | Turn       | High       | 11.04515, 77.73621 | 15 KM/Hr    |
| 30 | Turn       | High       | 10.99026, 77.72511 | 15 KM/Hr    |
| 31 | Turn       | High       | 10.98974, 77.72388 | 15 KM/Hr    |
| 32 | Blind Spot | Blind Spot | 10.96959, 77.71867 | 10 KM/Hr    |

### Emergency Locations

|   | type     | name                             | coordinates               | speed_limit | risk_level |
|---|----------|----------------------------------|---------------------------|-------------|------------|
| 0 | hospital | Tiruchengode, Goverment Hospital | 11.3903328,<br>77.8920627 | 30 km/h     | Medium     |

|    | type     | name                             | coordinates               | speed_limit | risk_level |
|----|----------|----------------------------------|---------------------------|-------------|------------|
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331,<br>77.8931963 | 30 km/h     | Medium     |
| 5  | clinic   | Kongu Nursing Home               | 11.3783065,<br>77.8961134 | 30 km/h     | Medium     |
| 6  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885,<br>77.8965774 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429,<br>77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426        | 30 km/h     | Medium     |
| 9  | hospital | Tirukumaran Hospitals            | 11.3782118,<br>77.8914933 | 30 km/h     | Medium     |
| 10 | hospital | Krishna Hospital, Namakkal       | 11.3754811,<br>77.8931817 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல்<br>நிலையம் | 11.357586,<br>77.8922539  | 30 km/h     | Medium     |
| 14 | hospital | Government Hospital, Elumathur   | 11.18533, 77.772496       | 30 km/h     | Medium     |
| 15 | hospital | Malar Hospital, Muthur           | 11.0438049,<br>77.7361156 | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                   | coordinates               | speed_limit | risk_level |
|----|-------------|----------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப்<br>பள்ளி        | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப்<br>பள்ளி       | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி<br>காய்கறி சந்தை | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution            | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                             | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |

## Route Photos of Risky Spots





**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.04637, 77.73761





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44022, 77.87546



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37868, 77.89498



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.37842, 77.89452





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.30869, 77.92911



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.30849, 77.92910





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24172, 77.87357



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.24000, 77.87363



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24053, 77.86902





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.24115, 77.86891



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.24122, 77.86883





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.24159, 77.86874**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.24170, 77.86842**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.24248, 77.86829**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.24257, 77.86776



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.24384, 77.86689





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24101, 77.86034



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24103, 77.86020



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.24200, 77.85882





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.24196, 77.85856



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.18856, 77.77110





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16404, 77.77379



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16377, 77.77421



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16354, 77.77435





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13664, 77.76524



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.04718, 77.73742





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.04515, 77.73621



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.99026, 77.72511



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.98974, 77.72388





**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 10.96959, 77.71867

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